

WASP-43b

R-2419

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_150124 · created 2026-08-21T23:29:51+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457046.81254 +/- 0.00086 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.051
Transit depth (Rp/Rs)^2	3.1 +/- 1.81 [%]
Orbital Inclination (inc)	77.86 +/- 2.86
Airmass coefficient 1 (a1)	1.043 +/- 0.068
Airmass coefficient 2 (a2)	-0.028 +/- 0.028
Scatter in the residuals of the lightcurve fit is	6.41 %
Best Comparison Star	#2 - [336, 192]
Optimal Method	PSF photometry
Transit Duration (day)	0.028 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.176 ± 0.051 vs 0.1594 (deviation 0.32σ)
Duration fit vs published	0.028 d vs 0.0512 d (deviation 1.21σ)
Mid-time O-C	1.7 min
Depth SNR	3.4
Residual scatter	6.41 %

Light curve

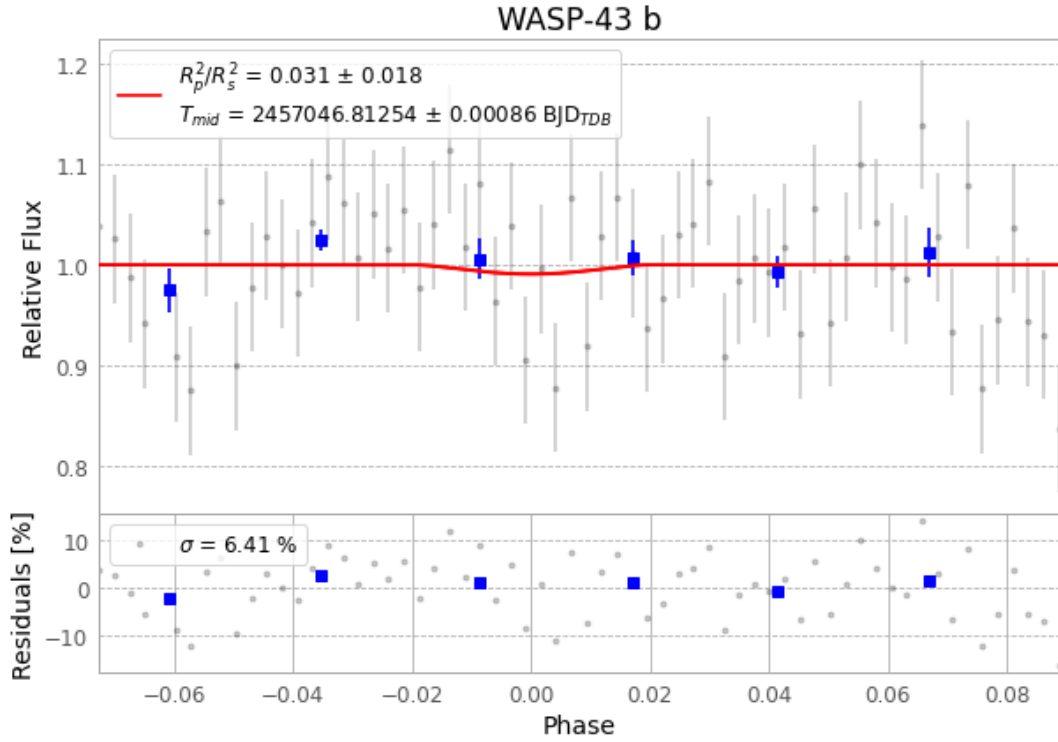


Figure 1 — FinalLightCurve_WASP-43 b_2015-01-23.png (SHA-256 14a5290e5407a398...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [356, 205], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 34242a78f08b8267...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43150124055712.FITS → WASP-43150124090612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-150124.log (6a892aa79bb0...), FinalLightCurve_WASP-43 b_2015-01-23.png (14a5290e5407...), FinalLightCurve_WASP-43 b_2015-01-23.csv (466cd5a064a9...)

Compute resources

Wall time 138.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 23:29Z.